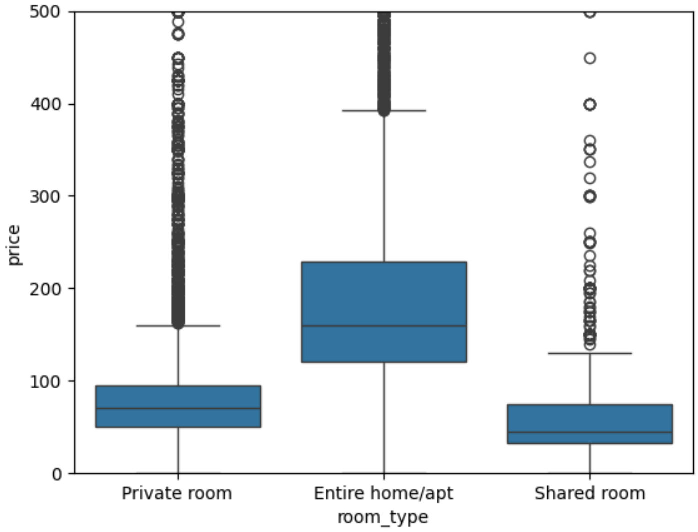
The dataset contained 48,895 rows and 16 columns, providing a large sample for pricing analysis and segmentation.

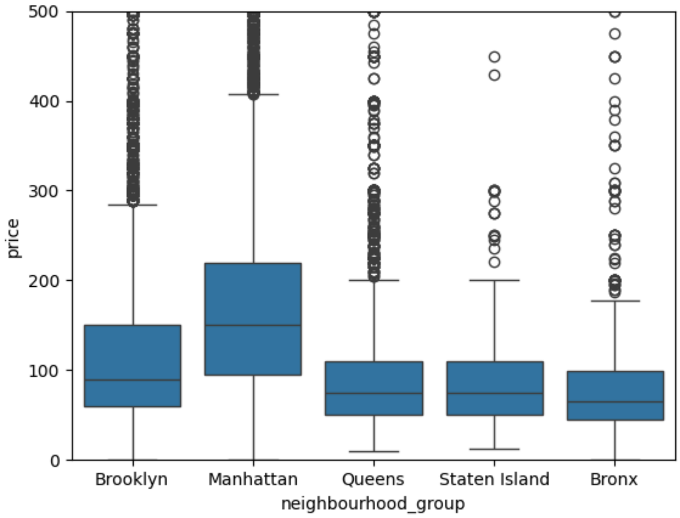
A large number of missing values were present in the review-related variables, suggesting that many listings have not been reviewed, rather than being random data quality issues.

The price distribution was highly right-skewed. The median price is $106.00 , the mean price is $152,00, but the maximum price is $10,000. This suggests that most listings are cheap, with a few outliers that increase the mean.

Price outliers were identified, with prices reaching $10,000, but the 99th percentile of prices is around $799. In addition, some listings had $0 as prices, which is unlikely and not useful to represent valid market prices. These records were removed during cleaning.



The room type boxplot shows that the listings with higher prices and the highest outliers are associated with the entire apartments, while the private room category has lower prices than apartments, but still contains many outliers. The lowest prices were in the shared room category, with fewer outliers.



The borough's boxplot shows that location is a major indicator of price. Manhattan has the highest median prices and the widest price dispersion, indicating a premium market compared with other boroughs. Brooklyn is the second most expensive borough, which could suggest that there is a strong demand but lower price levels than in Manhattan. The lowest prices are found in Queens, the Bronx, and Staten Island, which are the most affordable market segments.

Moving on to availability, 25% of listings have zero availability days, which could indicate inactive properties or hosts renting only occasionally. The median was 45 days, suggesting that many hosts are not continuously available throughout the year. The mean availability is 112 days, meaning that there are also listings with higher availability, suggesting professional hosts and the presence of investment properties.

DATA CLEANING

To improve data quality, the dataset was cleaned and prepared for regression and clustering models.

The missing values in reviews_per_month were filled with zero, while the rows with missing names or host_names were removed. Listings with prices equal to $0 were also excluded because they are unrealistic and would not be suitable for modelling.

Then, a log transformation was applied to reduce price skewness, creating a new variable called "log_price". The categorical variables, such as room type and neighbourhood, were numerically encoded for modelling purposes.

To handle the outliers in the minimum_nights variable, it was capped at the 99th percentile.

Final validation checks confirmed that there were no missing values in the modelling variables, and a clean dataset was produced for the next stage of the process.